Round 16 - Signed Bilinear Quartic Memory (SBQM)

One full-rank matrix state; long signed memory; genuine all-distinct quartic sector

[Derived design] Carry one symmetric dense matrix H_l in addition to the covariance C_l. Interpret the quartic memory as the fully symmetric bilinear tensor Q(H,C)=Sym(H tensor C). H is transported by the signed full weight map W (not W squared), while only its diagonal is refreshed at each ReLU from the already-shipped chain/V55 quartic diagonal. The implied tensor has non-zero (2,1,1) and (1,1,1,1) entries. Exact incremental cost: 2 ordinary 1024x1024 GEMMs per inter-layer map = 30 u over 15 maps, plus O(n^2) work; estimated extra metered operations < 200.

[Execution status] I attempted the requested stored-data run. The authorized FOYSAL data host is offline, and the raw teacher/capture NPZ files are not mounted or present in the ChatGPT Library. Therefore no fresh correlation is reported as measured. Section 5 gives the exact ceiling script/test to run as soon as the host is online. All reported correlations below are quoted from the prompt, not re-measured here.

1. State, implied quartic tensor, and readout

[Derived] State. At pre-activation layer l carry H_l in R^(n x n), symmetric and full-rank. C_l is the covariance the chain already carries. Define the implied fourth cumulant

$$Q(H,C)_{abcd} = 1/2 \; [\; H_{ab} C_{cd} + C_{ab} H_{cd} + H_{ac} C_{bd} + C_{ac} H_{bd} + H_{ad} C_{bc} + C_{ad} H_{bc} \;].$$

[Derived] Genuine all-distinct sector. For four distinct indices a,b,c,d the expression above is generically non-zero. It is therefore not a K22-only carrier, not diagonal-only transport, and not Sym(C tensor C) unless H happens to equal C.

[Derived] Slices/readout. For symmetric H and C:

$$K4_{iiii} = 3 \, H_{ii} C_{ii}$$

$$K31_{ij} = (3/2) \; (\; H_{ii} C_{ij} + C_{ii} H_{ij} \;)$$

$$K22_{ij} = (1/2) \; (\; H_{ii} C_{jj} + C_{ii} H_{jj} + 4 \, H_{ij} C_{ij} \;).$$

[Derived] Mean-closure input. Use q4_hat,l[i] = 3 H_l[ii] C_l[ii], then rescale this vector to the chain/V55 per-layer quartic level before feeding the existing univariate Edgeworth mean closure. Scalar rescaling does not change pattern correlation.

2. Birth/refresh and transport

[Derived] Linear transport. For the signed linearized ReLU map D_l = diag(Phi(mu_l/sigma_l)) and z_(l+1)=W_l^T h_l, transport the off-diagonal memory by

$$H_l^{carry} = D_l \, H_l \, D_l, \qquad H_{(l+1)}^{pre} = W_l^T \, H_l^{post} \, W_l.$$

[Derived] ReLU refresh. Do not regenerate the full matrix. Keep every off-diagonal element of H_l^carry and overwrite only the diagonal so that the local quartic diagonal equals the already-shipped chain/V55 value q4_chain,l on the current post-ReLU variance v_post,l:

$$H_l^{post}[i,j] = H_l^{carry}[i,j] \;\; (i \neq j), \qquad H_l^{post}[i,i] = q4_{chain,l}[i] \; / \; (3 \, v_{post,l}[i]).$$

[Inference] This separates jobs: the diagonal refresh supplies fresh local amplitude; the off-diagonal H carries the signed orientation for many layers. Dense W congruence continuously converts old off-diagonal memory into future diagonals, so a seed from six or more layers earlier can still affect q4_hat without storing a quartic tensor.

[Why the prior kills are too narrow] The dead pair-response/previous-feature carriers propagate K22-like or per-neuron objects through even W-squared maps and therefore do not preserve a genuine all-distinct signed sector. SBQM transports a full signed matrix through W^T H W and only forms a quartic object through the bilinear completion with C. The dead Sym(C tensor C)/scale-mixture case is the special case H proportional to C; SBQM does not impose that relation.

3. Exact incremental cost at n = 1024

operation	real shape	count/network	cost
signed H transport: T = H W	1024x1024 by 1024x1024	15	15 u
signed H transport: H_next = W^T T	1024x1024 by 1024x1024	15	15 u
D H D + diagonal refresh	elementwise 1024x1024	15	30 n^2 + O(n) FLOPs
q4 readout + level rescale	vectors length 1024	16	O(16 n) FLOPs
TOTAL dense square products	-	30	30 u

[Derived] FLOPs. 30 x 2^31 = 64,424,509,440 FLOPs = 30 u = 0.029296875 of the 2^41 budget. The DHD/refresh work is about 30 n^2 multiplies over the network, approximately 3.15e7 scalar multiplies, or about 0.0147 u if metered arithmetically.

[Estimated] Metered operations. Using ordinary GEMM, there are 30 large matmul calls. With each layer implemented as a handful of batched elementwise calls, the extra operation count should be well below 200 and therefore below the 1,000-op requirement. This is an estimate because flopscope call fusion is implementation-dependent.

4. Why the ceiling could be >= 0.7

- **[Quoted]** The exact linear carry of the true full K4 tensor reaches pattern correlation 0.77-0.80 at L12-L16. SBQM is explicitly a carried state and uses the same signed W congruence, rather than rebuilding the pattern from even per-neuron statistics.
- **[Quoted]** A six-layer memory horizon already gives pattern correlation about 0.77, while the earlier young K4 carrier only retained 2-3 layers. SBQM has no horizon cutoff: an off-diagonal entry can survive all remaining layers.
- **[Derived]** The tensor Q(H,C) contains K211 and K1111 entries automatically. Its diagonal after future dense mixing therefore depends on products of four distinct weight entries; this is the structural content missing from the memoryless chain.
- **[Derived]** The state is full-rank spatially. There is no Tucker/TT/CP/sketch truncation and no assumption that the Hadamard products are low-rank.
- **[Estimated hypothesis]** The only substantive ceiling question is whether one full signed bilinear mode H x C is expressive enough. If it is, the 30-u implementation is already within the requested cost/operation envelope. If not, a second independent H mode would roughly double transport cost to 60 u and should not be pursued unless the one-mode ceiling is very close.

5. ONE falsifier - teacher-forced unfactorised ceiling

[Derived test] Use the stored post-ReLU teachers at L11, L14 and L15 on MLP #0 (and the corresponding available synthetic layers). At each source layer, fit the best full symmetric H to the teacher K31 and K22 slices with C fixed to the teacher covariance; do not use the next layer q4 in the fit. Then transport H exactly once with the actual W to the next pre-activation layer and read out q4_pred = 3 diag(H_next) diag(C_next). Correlation is evaluated on decisive neurons |mu/sigma| < 1.5.

[Derived fitting equations] If the source-layer q4 diagonal is used only to set H_ii for the most favorable transport ceiling, set H_ii=q4_i/(3 C_ii). For each i<j choose H_ij by least squares over the three available slice equations:

$$a1 = 1.5\,C_{ii},\quad b1 = K31_{ij} - 1.5\,H_{ii}\,C_{ij}$$
$$a2 = 1.5\,C_{jj},\quad b2 = K31_{ji} - 1.5\,H_{jj}\,C_{ij}$$
$$a3 = 2\,C_{ij},\quad b3 = K22_{ij} - 0.5(H_{ii}\,C_{jj} + C_{ii}\,H_{jj})$$
$$H_{ij} = (a1\,b1 + a2\,b2 + a3\,b3) / (a1^2 + a2^2 + a3^2 + \mathrm{eps}).$$

[Derived one-step transport] At a post-ReLU source layer use H_next = W_l^T H_post W_l. No D factor is needed because the source teacher is already post-ReLU. Compare q4_pred with the stored next pre-activation teacher. This isolates the representational/transport ceiling and does not test the cheap diagonal-refresh closure yet.

[PASS/KILL GATE] PASS the state only if teacher-forced one-step pattern correlation is >= 0.70 on every available deep target among L12-L16 on both networks (or, if a target is unavailable on the synthetic teacher grid, on every available deep target). If the oracle-fitted one-matrix state misses 0.70, kill SBQM immediately: the cheap diagonal-refresh version cannot rescue an insufficient state family. If it passes, the next experiment is the 30-u free run from layer 1 using the diagonal refresh in Section 2.

[Execution attempt, measured status] No numerical ceiling was produced in this session: the FOYSAL Desktop Commander device reports Offline, and searches of the mounted container/Library found no raw k4teach/r7_cap NPZ files. Therefore the PDF intentionally contains no fabricated "measured" correlation.

[Implementation note] The ceiling fit is vectorizable: H_ii is O(n); all H_ij values are elementwise n x n operations; one-step transport is exactly two n^3 GEMMs per tested layer. The fit itself is offline analysis and is not part of the estimator FLOP ledger.